A FINITE CENTRAL SKEW FIELD HAS A MAXIMAL SEPARABLE SUBFIELD

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced from the author's Brauer groups chapter, brauer.tex, fixed at Stacks Project commit a04446e57ec1fbc252a871afcec7752fb2807b14. The source supplied here has the native preamble inlined before the chapter, so its line numbers differ from the separate chapter source.

Editorial note. The author writes '(details omitted)' for the coordinate-polynomial expansion of the q-th power map. This local compression is retained verbatim; no proof was supplied by the editor.

Editorial note. The named standard prerequisites are given below. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Editorial note. The preamble spans are copied from sources/stacks-brauer/pinned-original-source.tex. This revision corrects two chapter-name errors in the previous editorial notes; it does not change the author's mathematical statement or proof.

Allowed standard prerequisites (editorial).

- A maximal subfield splits the algebra. Same chapter, Lemma lemma-maximal-subfield-splits. Supplies that a finite central skew field is split by a maximal subfield, which turns the second assertion of the statement into the first.
- The centralizer of a separable subfield. Same chapter, the discussion of centralizers of subfields of a central skew field. Supplies that the centralizer of a separable subfield again has that subfield as its center, which is what allows the author to replace the ground field and reduce to finding one separable element.
- Inseparable elements and minimal polynomials. Fields, Section fields-section-algebraic. Supplies that in positive characteristic an element whose minimal polynomial is not separable has a minimal polynomial of the form a p-power variable minus a constant.
- Base change and the algebraic closure case. Same chapter, Lemma lemma-base-change and Lemma lemma-brauer-algebraically-closed. Supply that extending scalars to the algebraic closure turns the skew field into a matrix algebra, which produces the element whose p-th power is not central.

Source reference. Original lemma 0750: sources/stacks-brauer/pinned-original-source.tex, source lines 896–896. Navigation pointer invoked at proof line 935; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 037H: sources/stacks-fields/pinned-original-source.tex, source lines 3909–3909. Navigation pointer invoked at proof line 952; the counted

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span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074H: sources/stacks-brauer/pinned-original-source.tex, source lines 565–565. Navigation pointer invoked at proof line 975; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074M: sources/stacks-brauer/pinned-original-source.tex, source lines 667–667. Navigation pointer invoked at proof line 975; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Consider a finite central skew field K over k. There exists a maximal subfield $k \subset k' \subset K$ which is separable over k. In particular, every Brauer class has a finite separable spitting field.*

Proof. Since every Brauer class is represented by a finite central skew field over k, we see that the second statement follows from the first by Lemma 0750.

To prove the first statement, suppose that we are given a separable subfield $k' \subset K$. Then the centralizer K' of k' in K has center k', and the problem reduces to finding a maximal subfield of K' separable over k'. Thus it suffices to prove, if $k \neq K$, that we can find an element $x \in K$, $x \notin k$ which is separable over k. This statement is clear in characteristic zero. Hence we may assume that k has characteristic $p > 0$. If the ground field k is finite then, the result is clear as well (because extensions of finite fields are always separable). Thus we may assume that k is an infinite field of positive characteristic.

To get a contradiction assume no element of K is separable over k. By the discussion in Fields, Section 037H this means the minimal polynomial of any $x \in K$ is of the form $T^q - a$ where q is a power of p and $a \in k$. Since it is clear that every element of K has a minimal polynomial of degree $\leq \dim_k(K)$ we conclude that there exists a fixed p-power q such that $x^q \in k$ for all $x \in K$.

Consider the map

$$(-)^q : K \longrightarrow K$$

and write it out in terms of a k-basis $\{a_1, \ldots, a_n\}$ of K with $a_1 = 1$. So

$$(\sum x_i a_i)^q = \sum f_i(x_1, \ldots, x_n) a_i.$$

Since multiplication on K is k-bilinear we see that each f_i is a polynomial in $x_1, \ldots, x_n$ (details omitted). The choice of q above and the fact that k is infinite shows that f_i is identically zero for $i \geq 2$. Hence we see that it remains zero on extending k to its algebraic closure $\overline{k}$. But the algebra $K \otimes_k \overline{k}$ is a matrix algebra (for example by Lemmas 074H and 074M), which implies there are some elements whose qth power is not central (e.g., e_{11}). This is the desired contradiction. $\square$